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To: elias@lcad.inf.ufes.br

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Dear Dr. Elias Oliveira,

We are happy to inform you that the regular paper you have submitted to KDIR, with number 76, entitled ''Explaining the Judges' Decisions Criteria'', has been accepted as a Short Paper. Papers accepted as Short Paper are assigned a 8-page limit in the conference proceedings.

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Should you have any other question please don't hesitate to contact the secretariat.

Best regards,
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Explaining the Judges' Decisions Criteria

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Abstract: The identification of named entities in free text is a foundational research area for building intelligent systems in text and document mining. These textual elements allow us to evaluate the reasoning expressed by document authors. In a judicial decision, for example, by identifying time-related entities, an intelligent system can assess and verify whether a sentence issued by a justice agent falls within socially agreed-upon statistical parameters. In this study, 769 judicial decisions from the São Paulo court were evaluated. Our experiments compared the extreme time-value sentences against those with the lowest sentence, for instance, to infer the expressions that justified and have explained their values. The results revealed differences in sentence severity among robbery, drug trafficking, and theft, as well as in how judges cluster based on their sentencing behavior. The study also highlights anomalies in sentencing and links them to specific textual justifications, demonstrating how judges' decisions can reflect both legal criteria and subjective biases. [...] *In a lawsuit the first to speak seems right, until someone comes forward and cross-examines.* (Proverbs 18:17)

1 INTRODUCTION

Information retrieval tools have significantly improved over the last decade. Companies like Google, Yahoo, and Bing provide powerful search engines that enable near-instantaneous access to vast amounts of documents worldwide. Nevertheless, we need to further improve these tools to provide the user with better search engine for more complex meta-indexing (Izo et al., 2021). Inasmuch as similar tools are still lacking in many information systems used by the Brazilian justice department. Integrating such tools into local institutional systems could both enhance efficiency by enabling precise and rapid retrieval of information from extensive document archives (Lima et al., 2024) and allow one to perform complex inferences on hitherto statistical parameters.

The judge's decision-making landscape must be minimally justified in their decision document. Therefore, to apply a sentence s_i , the judge must justify it by presenting facts $\mathcal{F}$, in other words, we must find these elements, $F_{acts} \rightarrow s_i$, explicitly registered in the decision. $\mathcal{F}$ can be either the defendant's past criminal history f_1 , or the defendant's ties to the community f_2 , or the defendant's employment history f_3 , or any other fact f_n , or all together, one may add to the decision's description. Therefore, $F_{acts} = \{f_1, f_2, \dots, f_n\}$, where $F_{acts} \subset \mathcal{F}$, are the explanation a judge gives to charge someone with a s_i penalty.

Judges can also adopt the strategy of comparing two, or more cases, to set up their decision. In this case, a threshold is used to say that F_i and F_j – where $F_i, F_j \subset \mathcal{F}$ – are somehow similar to induce the same sentence s_i . By doing so, judges' decisions become as rational and clear as mathematics, aligning with society's expectation that judges remain impartial and dispassionate (Maroney, 2011). Unfortunately, some judges do not decide entirely on the record. They add to the judgment decision their emotional feeling and

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sympathy instead of the logic and strict description of the law. Other studies, discussed by Rachlinski and Wistrich (2017), demonstrate that specific religious values lead judges to decide certain kinds of cases differently. Also, Rachlinski and Wistrich (2017) examined various cognitive biases that may influence judges when issuing their verdicts. Some are: (a) demographic characteristics of judges. (b) relying heavily on intuitive reasoning and not on facts, or even worse, relying on facts outside the judicial records. In Brazil, we may include the social class of the defendant as a bias factor on the judicial decisions. In theory, judges' beliefs should not override the original agreement expressed by the people in the ordinary laws established under the Brazilian Constitution.

The judge's decision is the final administrative act issued by the judicial system referring to facts disputed by parts. Having this in mind, these documents can be valuable instruments to assess the performance of the system. However, to achieve this goal, we need to build a better archive system to increase the accessibility of these documents by attaching them to new access points. Today's accesses are based mainly on the number of processes, the name of the parts, the subject type in dispute, and keywords.

The extraction of the aforementioned metadata may be straightforward using pattern recognition strategies. In this work, we go beyond that by extracting other information within these documents, such as the names of other people mentioned, the dates of the referenced events, and the time prescribed by the court. More complex pieces of information require Natural Language Processing (NLP) (Manning and Schuetze, 1999). To handle these more complex extractions, this work incorporates prompt engineering techniques, which have proven effective in guiding natural language models. This approach involves the careful formulation of instructions for Large Language Models (LLM)s, enabling a more efficient use of their capabilities. Understanding how to construct, refine, and strategically apply prompts is essential to maximizing these models' performance. Well-crafted prompts allow the model to recognize subtle patterns, such as temporal relationships and specific entities (Schulhoff1 et al., 2024). Thus, prompt engineering serves as a bridge between expert knowledge and artificial intelligence, expanding the practical application potential of these models.

The contributions of this work are threefold. First, we make publicly available a large dataset of court decisions regarding the crimes discussed here. Second, we organize these decisions into popular software frequently used by archivists, such as *AtoM*¹. Finally,

¹<https://www.accesstomemory.org/>

we analyze 769 decisions from 290 different equivalents to USA district judges. We attempted to explain their statistical similarities and discrepancies by extracting factual elements from their decision documents.

To present our proposal, we consider only four types of crime: 1) Simple Theft, 2) Qualified Theft, 3) Robbery and 4) Drug Trafficking. We formed an archive of 769 decisions from that issued by the *São Paulo Court of Appeal* – (TJSP) in 2022 only.

This article is structured as follows. In Section 2, we discuss some recent works that will serve as a basis for comparison with our proposal. In Section 3, we present our proposal with an emphasis on the efficient use of computational resources and artificial intelligence techniques aimed at the extraction and structuring of legal data. In Section 4, we discuss the experiments and results. Finally, we present our conclusions in Section 5.

2 LITERATURE REVIEW

Advances in artificial intelligence and data analytics have enabled computers to predict judicial decisions by analyzing patterns in case law, judges' rulings, and legal reasoning.

Regarding judicial decision-making and the application of technologies, the seminal Lawlor's approach, developed by legal scholar Reed C. Lawlor, arose in the 1960s. His work applied early computational logic to legal reasoning, using syllogistic structures to model how judges interpret facts and legal rules. Lawlor's method broke down judicial decisions into discrete components - facts, legal principles, and outcomes - to identify predictable decision pathways. From 70s to 90s, there were proposals to combine the automation of legal reasoning and expert systems, such as: TAXMAN (McCarthy, 1977) and FINDER (Tyree et al., 1987). Later, a computational breakthrough was introduced by Susskind (1998), and Ashley (1990) proposed a rule-based and case-based reasoning paradigm, respectively. *These theoretical debates set the stage for computational models aiming to replicate or predict judicial reasoning.*

A paradigm shift from symbolic AI to data-driven methods was proposed by Bench-Capon and Sartor (2003), emphasizing the value-driven reasoning in AI systems, and Katz et al. (2017) applying techniques such as machine learning, considering data-based decision-making but ignoring some *contextual factors*.

Recent advances have pointed out on interdisciplinary innovations, for instance, NLP and Bench-

marking Chalkidis et al. (2023), with *LexGLUE* dataset standardizing legal NLP tasks; and topics as ethical risks (Sourdin, 2021), such as the ethical concerns about replacing judges with algorithms; biases and hallucinations in automatic judicial decision making (Medvedeva et al. (2020): how evaluates biases in AI predictions of human rights cases, and explainability (how adapt AI for different legal systems).

In recent years, several projects have been suggested to study how technology can be used in courts. One of these projects is called ADELE. The ADELE project tried to create a plan for using technology to understand legal data. It focused on court decisions to help analyze and process legal data more effectively. The main goals were to enhance legal data analysis, improve accessibility to case law insights, and provide a structured approach to interpreting judicial decisions in these domains. The approaches applied were Natural Language Processing (NLP) and machine learning (ML) to analyze court decisions in Italian and Bulgarian case law, focusing on Intellectual Property Rights (IPR) and Value Added Tax (VAT). Key methods included text preprocessing, named entity recognition (NER), and topic modeling to extract legal concepts, classify cases, and identify patterns. The project also utilized predictive analytics and knowledge graphs to map legal relationships and precedents, enabling structured analysis of judicial decisions.

While these approaches offer valuable insights for legal practitioners and policymakers, challenges remain, such as ensuring algorithmic transparency and addressing biases in training data. Nevertheless, the integration of computational methods into judicial analysis represents a significant evolution in legal research, bridging traditional doctrinal analysis with data-driven forecasting.

As regards the legal nature of judicial decision, its definition varies across legal traditions, for instance, in Legal Positivism theory, a judicial decision is "the application of pre-existing rules to facts, deriving authority from the legal system's hierarchy of norms." (Hart, 1961); in Legal Realism theory, judicial decisions are influenced by contextual factors (e.g., judge's worldview, societal values) beyond formal rules (Fuller, 1958); in Legal Interpretivism, decisions encompass principles (e.g., justice, fairness) alongside rules, constructing a coherent moral narrative (Dworkin, 1977), just to cite some legal doctrines. *These theories reveal tensions between rule-bound formalism and discretionary judgment, shaping computational modeling attempts.*

The Brazilian penal legal system is grounded in a combination of legal theories (e.g., legal positivism,

social-critical critiques) that shape its substantive (*Direito Penal*) and procedural (*Direito Processual Penal*) dimensions. In this work, the scope of the Brazilian theory of judicial decisions was limited to sentences, which are a final court ruling that resolves the merits of a case. Interlocutory decisions or provisional orders were excluded from the scope.

A *Judicial Sentence* is issued by a competent judge, who articulates the factual and legal basis for the ruling. In terms of communicative acts, judicial sentences are a subtype of institutional acts (Searle, 1995; Austin, 1975), i.e., an act requiring objective and subjective validity conditions. Under the Brazilian Code of Criminal Procedure (*Código de Processo Penal – CPP, Art. 381 to 383*), to be considered as a valid judicial sentence (*sentença penal*) must be signed and dated, and necessarily containing 1) the identification of all parties involved in the case, including their full names or, when such information is unavailable, any other details sufficient to establish their identity; 2) must present a concise description (*aka relatório*) of both the prosecution's accusations and the defense's arguments, outlining the key claims and counterarguments that frame the legal dispute; 3) the legal reasoning (*aka fundamentação*), which is the factual analysis, the subsumption of the fact to the criminal type (analyzing materiality, culpability, legal grounds). For example, the defendant's nocturnal entry into the victim's property, captured on CCTV, satisfies the aggravator of 'night-time' under Art. 155, §4º; 4) the dispositive part (*aka dispositivo*), containing the final ruling, a clear decision on conviction or acquittal, the penalty in case of convicted, and, in some cases, civil liability. Also, there are some differences between criminal type definitions as shown in Table 1.

The legal definitions and treatment of theft, robbery, and drug trafficking reveal significant differences between Brazil's civil law system and the United States' common law tradition. These distinctions reflect broader variations in legal philosophy, legislative structure, and societal priorities.

In Brazil, simple theft is defined under Article 155 of the Penal Code as the unlawful taking of movable property without consent and without violence, carrying a penalty of one to four years imprisonment. The U.S. approach, guided by state statutes and the Model Penal Code, focuses instead on the value of stolen goods, distinguishing between petty theft (minor offenses) and grand theft (more serious crimes), with penalties varying accordingly. Brazilian law further recognizes qualified theft, where aggravating factors such as breach of trust or nighttime commission increase the sentence, while U.S. law lacks an exact

counterpart, instead enhancing penalties based on victim vulnerability or stolen property value.

Robbery in Brazil, as outlined in Article 157 of the Penal Code, requires theft accompanied by violence or grave threat, punishable by four to ten years in prison. The U.S. adopts a broader definition, where any use of force or intimidation during theft constitutes robbery, with penalties ranging from ten years to life imprisonment, particularly in cases involving firearms. This reflects a stricter stance on crimes involving personal violence in the U.S. compared to Brazil’s more narrowly defined thresholds for aggravated theft.

Drug trafficking presents another area of contrast. Brazilian law (Law 11.343/2006) criminalizes unauthorized production, transport, or sale of illicit drugs, with sentences of five to fifteen years, while distinguishing between personal use and trafficking based on quantity. The U.S., under federal law (21 U.S.C. § 841), emphasizes intent to distribute, imposing severe penalties—often five to forty years—depending on drug type and quantity, with mandatory minimum sentences playing a prominent role.

These differences highlight how Brazil’s codified legal system provides precise definitions and fixed sentencing ranges, whereas the U.S. system’s flexibility allows for contextual adjustments based on judicial interpretation and legislative enhancements. Understanding these distinctions is crucial for comparative legal studies and cross-jurisdictional policy analysis.

3 THE METHODOLOGY

A judge’s decision statement contains numerous facts, F_{acts} , that may hold valuable information for someone in the future. However, it is impossible to predict *a priori* which words, complex textual structures, or other linguistic phenomena will be most useful to a researcher, citizen, or archivist as their access points. Therefore, we need enhanced tools and methodologies to efficiently adapt archival description structures to meet users’ specific needs. The case study proposed in this work is the evaluation of the behavior of judges on sentencing. This information was not previously available to the information system for us to process and perform various statistical analyses. For that reason, we needed to extract them from the documents using modern artificial intelligence tools.

To protect the privacy of all parties involved in the analyzed judicial documents, we adopted a rigorous anonymization process. All personal information – including names of defendants, victims, wit-

nesses, and judges – was removed or replaced with coded identifiers before applying the clustering algorithms. Additionally, we ensured that the identifiers used do not allow reverse re-identification, even through cross-referencing analysis. This careful approach is especially important in grouping judges by sentencing patterns, where we take additional measures to prevent any risk of individual exposure.

3.1 The prompting engineering strategy

To construct the database, structured information was automatically extracted from judicial rulings, including case number, category, district, court division, presiding judge, imposed sentences (base, provisional, and final), as well as judicial circumstances and aggravating or mitigating factors such as recidivism, use of weapons, attempt, voluntary confession, and characteristics of both the defendant and the victim. The extraction process was carried out with the assistance of the Qwen 2.5-32B-Instruct language model (Qwen et al., 2025), an open-source model with 32 billion parameters, run locally. The choice of this dense version of the model ensured greater control over processing, preservation of the confidentiality of judicial decisions, and advanced performance in tasks involving legal language understanding and structured data handling. A prompt was developed to traverse all decisions, identifying sentences and correctly linking them to the respective individuals, as some rulings involve multiple defendants. This step was essential to properly structure the data, enabling subsequent analyses of judicial behavior in sentencing decisions.

The prompt followed a zero-shot approach, in which the model is guided solely by natural language instructions without being provided with prior input-output examples (Boyina et al., 2024). This technique allows for quick adaptation to new texts and tasks, although it may face limitations in more complex or ambiguous contexts, as is often the case with judicial rulings. For this reason, few-shot variations were also tested, incorporating concrete examples to guide the model regarding the expected format and the type of information to be extracted (Huang and Wang, 2025). The combination of these approaches proved effective in increasing the accuracy and consistency of the extraction process, contributing to the creation of a reliable and reusable dataset for empirical analyses of the criminal justice system.

Table 1: Comparative analysis of theft, robbery, and drug trafficking: Brazil vs. U.S.

Crime Type	Brazilian Law	U.S. Law	Key Distinctions
Simple Theft	Definition: Unlawful taking of movable property without violence or consent (Art. 155, CP). Penalty: 1–4 years + fine.	Definition: Unlawful taking with intent to deprive the owner permanently (no force). Penalty: Petty theft (≤ 6 months) to grand theft (1–3 years, state-dependent).	Brazil focuses on property type (movable), U.S. prioritizes property value thresholds.
Qualified Theft	Definition: Theft + codified aggravators (e.g., breach of trust, nighttime; Art. 155, §4°). Penalty: 2–8 years.	No standalone category; aggravators (e.g., victim age, high value) elevate to felony theft.	Brazil’s aggravators are explicit in law, U.S. uses contextual enhancements.
Robbery	Definition: Theft with violence/grave threat (Art. 157, CP). Penalty: 4–10 years (increased if armed).	Definition: Theft by force/intimidation (e.g., armed robbery; 18 U.S.C. § 1951). Penalty: 10 years–life (varies by state).	Brazil requires grave threat; U.S. accepts any force/intimidation.
Drug Trafficking	Definition: Unauthorized production/sale of drugs (Law 11.343/2006, Art. 33). Penalty: 5–15 years.	Definition: Manufacture/distribution with intent to distribute (21 U.S.C. § 841). Penalty: 5–40 years (mandatory minimums apply).	Brazil distinguishes trafficking from personal use by quantity. U.S. focuses on proven intent.

4 EXPERIMENTS AND RESULTS

The experiments carried out in this work show the behavior of a set of district judges from São Paulo, the biggest city in Brazil, with regard of four different type of crimes (see Section 2). Initially, we evaluated the best statistical distribution to characterize the Simple Theft sample of 99 sentences. In Table 2, we show the statistics for these sample sentences issued in our dataset for that type of crime.

Table 2: The simple theft statistics

	Simple Theft			
	Base Penalty		Final Penalty	
	year	month	year	month
minimum	1.00	12.00	0.33	4.00
average	1.20	14.37	1.16	13.96
σ	0.33	3.99	0.58	6.95
median	1.17	14.00	1.17	14.00
maximum	3.00	36.00	5.00	60.00

Based on the facts presented by the state’s attorney and the defendant, the judge may determine a *base penalty* in their decision (refer to the first and second columns in Table 2). This base penalty serves as the starting point, while the *final penalty* (see the third and fourth columns) results from adjustments – either increases or reductions – based on the evaluation of previously described factors. The values in the second column indicate that the minimum sentence is one year (365 days), while the maximum is three years. The average lies close to the center of this in-

terval. Adding further statistical insight, the standard deviation ($\sigma = 0.33$) is less than one year, indicating that for this type of crime, the variation around the average sentencing is relatively small. The third column shows similar figures but in months, considering all months with 30 days. These are two usual ways for professionals in the justice domain to deal with sentences. Above all, one can notice that all the median figures among year and month columns are very similar to the average values, suggesting a symmetric normal-curve.

Besides the previous deterministic experiments, we also carried out a different approach to estimate the same value discussed in Table 2 by applying the Probabilistic Programming (PP) technique with Stan language (Carpenter et al., 2017). PP is an approximation strategy, therefore the values of average and σ are not exactly the same of that calculated by the deterministic strategies in Table 2. Currently built within many modern languages, such as Python and R, PP uses an efficient engine to sample and re-sample to find the converging parameters to find a statistical distribution model for a given problem (Gelman et al., 2013). Nevertheless, we show the convergence of the approximated averages, for years, using a set of different samples in Figure 1.

The convergence in Figure 1 pointed out the average estimated by our PP approach. The approximated average given by the PP tool is 1.16, and the standard deviation is 0.59 at the level of the 65 sample items, much similar to that provided by the deterministic procedure in Table 2. Besides, note that the cred-

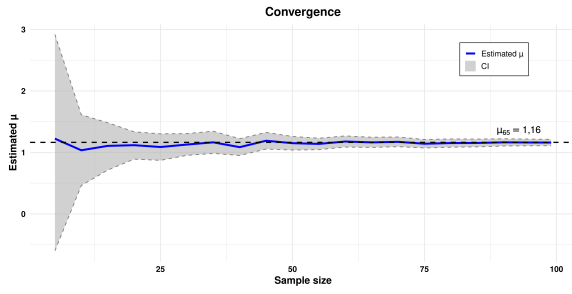


Figure 1: Convergence of years average for Simple Theft

ible interval shown in Figure 1 by the light gray area around the convergence line is narrowing as we increase the sample. This straight light gray area tells us we do not need a larger sample than 99 random sentences to conclude the average is 1.16 with 95% certainty.

The following Table 3 up to Table 6 present statistics for all four crimes considered in this study, each with a sufficient sample size to ensure that the average falls within a 95% credible interval.

Table 3: The simple theft statistics

Simple Theft – 276 sample					
	Base Penalty		Final Penalty		
	year	month	year	month	
minimum	0.29	3.50	0.22	2.67	
average	1.62	19.48	1.68	20.16	
σ	0.69	8.31	1.00	12.03	
median	1.25	15.00	1.33	16.00	
maximum	4.67	56.00	5.83	70.00	

Table 4: The qualified theft statistics

Qualified Theft – 174 sample					
	Base Penalty		Final Penalty		
	year	month	year	month	
minimum	1.00	12.00	0.67	8.00	
average	2.18	26.15	2.31	27.68	
σ	0.65	7.78	1.13	13.59	
median	2.33	28.00	2.04	24.43	
maximum	4.75	57.00	7.58	91.00	

Each table describes a model to explain mathematically its respective crime sentencing behavior by the population of judges of that district. With these *prior models*, we can compare and contrast judicial patterns across different districts and states in Brazil. In addition, this framework allows for discussions on whether these models reflect local culture and social norms or whether interventions might be necessary to influence or redirect judicial behavior.

A particularly interesting statistical analysis in this context is the analysis of variance (ANOVA),

Table 5: The robbery statistics

Robbery – 149 sample					
	Base Penalty		Final Penalty		
	year	month	year	month	
minimum	1.00	12.00	0.58	7.00	
average	4.57	54.82	7.52	90.25	
σ	0.78	9.35	4.24	50.90	
median	4.67	56.00	6.67	80.00	
maximum	6.67	80.00	36.00	432.00	

Table 6: The drug trafficking statistics

Drug Trafficking – 170 sample					
	Base Penalty		Final Penalty		
	year	month	year	month	
minimum	2.00	24.00	1.62	19.43	
average	5.48	65.77	4.63	55.55	
σ	0.87	10.41	2.47	29.69	
median	5.00	60.00	1.33	16.00	
maximum	8.00	96.00	9.72	116.67	

which revealed significant differences in the average sentences among the crimes evaluated (simple theft, qualified theft, robbery, and drug trafficking), with an F distribution value of 144.14 and $p < 0.0001$. This result shows that at least one group differs from the others. Tukey's *post hoc* test revealed that sentences for robbery are significantly higher than for the other crimes, and that drug trafficking also has higher average sentences compared to both simple and qualified theft. The only comparison without a significant difference was between simple theft and qualified theft, indicating that, in terms of average sentencing, these two types of theft are treated similarly in the dataset analyzed.

These results indicate a clear gradation in the severity of sentences, with robbery receiving the highest penalties, followed by drug trafficking, and finally theft-related offenses. Robbery stands out not only for having higher average sentences but also for greater variability, including extreme cases with very long sentences. Drug trafficking also shows a high median and greater dispersion compared to the theft categories. In contrast, simple theft and qualified theft exhibit similar patterns, with close medians and lower variability, reinforcing the lack of significant difference between them identified in the statistical analysis.

Note that within this dataset, there are 290 judges who, through their sentencing behavior, contributed to the models described by the parameters in the previous tables. In Figure 2, three clusters were defined with the aim of identifying patterns of similarity in sentencing, considering only those judges who sen-

tenced this type of crime. The choice of the number of clusters was guided by quantitative criteria, extracted during the automated clustering process. The K-Means algorithm was used, with its configuration driven by multiple internal validation indices, such as the Silhouette Score (SS), the Calinski-Harabasz index (CHS), and the Sum of Squared Errors (SSE). The combination of these indicators allowed identifying the three-cluster configuration as the most consistent in terms of group separation and internal cohesion of the data. Thus, the resulting structure reflects clusters based on objective quality metrics.

In Figure 2, the $N(\mu_{ST_l}, \sigma_{ST_l}) = N_{ST_l}(1.08, 0.37)$ normal distribution is a model to describe the first group of judges on the left. On the right, $N(\mu_{ST_r}, \sigma_{ST_r}) = N_{ST_r}(4.72, 0.58)$ explains the second group. Each model may account for the sentencing patterns observed among a particular group of judges in relation to this type of crime. The range of values of sentencing by these judges shows, hopefully, how different the circumstances faced by these judges for sentencing they did. Nonetheless, we are not yet looking into their sentencing text to combine the implication of their decision text and the time assigned to the sentence. We expect that the similar decision text should lead to the same time assigned to the sentence. These models also allow us to identify an unknown judge by one of these patterns by testing the judge characteristics under the null hypothesis, to a previous known $N_{ST}(\mu, \sigma)$. To build the models in Figure 2, we clustered all sentences' values in years and fractions because of months and days. That resulted in 51 decisions in the left cluster and 7 in the right cluster out of 276 decisions in total of this type of crime. The other sentences are in the clusters between these extreme ones.

So far, we illustrated our constructed mathematical structure only with $N_{ST}(\mu, \sigma)$ models, for simple theft. However, we have also built similar structures for: (a) $N_{QT}(\mu, \sigma)$, qualified theft; (b) $N_R(\mu, \sigma)$, robbery; and (c) $N_T(\mu, \sigma)$, drug trafficking.

The devised structures allow us to quickly search for the seven sentences, but one is overlapping another, placed on the right in Figure 2 as potential outliers sentences of this type of crime. We can also identify the judge, or judges, who have issued the court sentence in the $N_{ST_r}(\mu, \sigma)$ cases. Only six different judges issued those sentences, as shown in Figure 3. Among them, other twenty judges also issued sentences in different clusters. This unveils a very particular pattern of sentencing of these judges, which we can now look into the reasons why that occurred.

Similarly, we identified some outlier sentences for qualified theft in Figure 3. Using the same quantity of

four clusters previously applied to simple theft, four judges handed down sentences in the highest time-value group, while the remaining twenty were distributed among the other clusters.

To sum up, although the judges' decision texts differed from one another – explaining the variation in sentencing durations – some decisions appeared out of place or inconsistent with their respective clusters, as depicted in Figure 4.

We note that sentence 0, currently in Cluster 2, shares 95% similarity with Cluster 2. Similarly, sentence 1 shows 99% similarity with the same cluster. This suggests that both sentences could reasonably belong to Cluster 1, but are assigned to Cluster 2 due to their penalty values. Likewise, sentences 4 and 5 show strong similarity to those sentences in Cluster 1 – 100% and 97%, respectively. These sentences are considered outliers for this type of crime. Similar cases are also found on another type of crimes.

5 CONCLUSION

This work proposes the use of LLMs to extract complex information from legal texts, optimizing both archival description and the progress of legal proceedings through automation and semantic analysis. Unlike traditional approaches that focus only on elements such as case numbers or involved parties, our method analyzes the full text to identify named entities, allowing inferences about judges' behavior in their decisions.

The analysis focused on the duration of sentences to identify behavioral patterns, particularly in extreme cases. Empirical studies (Rachlinski and Wistrich, 2017) show that judicial decisions may be influenced by subjective factors—such as stereotypes or emotional states—rather than strictly legal reasoning.

One of the key findings is the connection between Figures 2, 3, and 4, all related to the crime of simple theft. Figure 2 presents the distribution of sentences across three distinct clusters, based on duration. Figure 3 retains the same sentences, now grouped by the judges who issued them, with colors preserved to represent the same clusters as in Figure 2. Figure 4, in turn, focuses on the sentences originally classified under cluster 2, redistributing them among the same three clusters—based on the textual similarity of each sentence to the centroids of these groups—while maintaining the same colors for comparison. This semantic comparison allows the user not only to verify thematic compatibility between the sentence and the reference group, but also to justify the inferred sentence based on observed linguistic patterns. The anal-

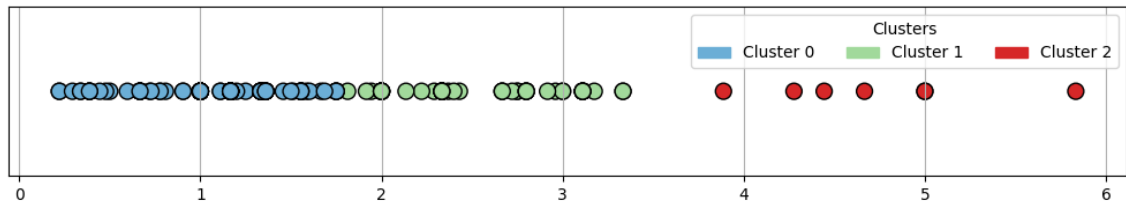


Figure 2: Clustering judges based on their sentencing to time – Simple Theft

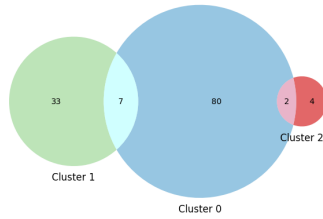


Figure 3: Simple Theft

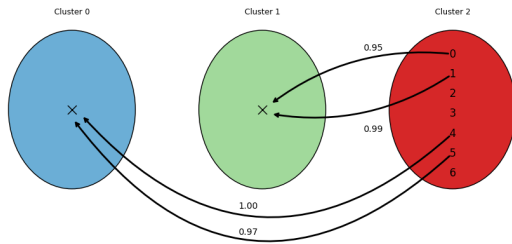


Figure 4: Clusters according to their texts

ysis reveals that some decisions, although grouped by sentence length, are linguistically closer to the discourses of other clusters. This suggests inconsistencies in reasoning and possible influence of extralegal criteria in sentencing, even in legally homogeneous cases.

The proposed tool proved effective in detecting linguistic patterns and behavioral anomalies. However, discrepancies were observed between the terms identified automatically and those recognized by legal experts. As future work, we propose the development of ontologies to define metadata types and synonyms, the segmentation of clusters into more homogeneous subgroups, and the filtering of sentences with low semantic cohesion. We also suggest using embeddings or legal dictionaries to normalize linguistic variations and reduce noise in the analyses.

Acknowledgement

The authors thanks José Jesus-Filho and Julio Trecenti for their support and for providing the software that enabled the collection and organization of data from the São Paulo Court of Justice (Jesus-Filho and Trecenti, 2020).

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KDIR 2025 - Paper #76

Paper Title: *Explaining the Judges' Decisions Criteria*

Reviewer #1

General Assessment (Please assign scores using the following criteria (1=weakest; 6=strongest))

Relevance (Paper fits one or more of the topic areas?): 5
Originality (Newness of the ideas expressed): 3
Technical Quality (Theoretical soundness/methodology): 3
Significance (Is the problem worth the given attention?): 5
Presentation (Structure/Length/English): 3
Overall Rating (Weighted value of above items): 3

Improvement Suggestions (for authors to consider in the camera-ready version. Additional detail in "Observations")

Abstract and Introduction are adequate? Yes
Needs more experimental results? Yes
Needs comparative evaluation? Yes
Improve critical discussion ? (validation): Yes
Figures are adequate ? (in number and quality): No
Conclusions/Future Work are convincing? No
References are up-to-date and appropriate? Yes
Paper formatting needs adjustment? Yes
Improve English? No

Detailed comments to authors, including aspects that must be improved in the camera-ready version of the paper:

Last weeks I've revised several papers whose titles were very promising, but the content was poor that I doubted the possibility to review something interesting and well written. The considered paper title reminded me the well-known and popular notion of XAI – Explainable Artificial Intelligence: the paradigm of providing understandable models for making decisions/predictions with AI tools. The context of the court issues seemed even more interesting, as it was almost sure, that some NLP models will be explained – models that will help judges to make more consistent (I consciously use the word "consistent" not the "fair") sentences. These were my expectations after reading the title and the abstract. We (the readers) will be given a tool that helps judges to focus on some important elements of the trial, evidences, circumstances extracted from similar past cases, for the better justification of the final judgement.

The expectations were rising while the Introduction, Literature review, and The named entities were read. Unfortunately, everything collapsed and became quite chaotic during the reading the rest of the paper.

Starting from the Table 3 we are given two goals of the analysis: the base and final penalty. We are not given why the "provisional" – mentioned in Section 3.1 – level was rejected from the analysis. But it is not so crucial – it just requires a small sentence in the text. The more important thing is a complete inconsistency in the further discussion. Just below the Table 3 we have a discussion about the "base penalty". I guess from the text that maximum = 3 is a data-driven value. Because Table 2 provides 1–4

years. And this is the first moment of doubts – the limits presented in the Tables 3–7 do not take into consideration the ability to sentence more years as it is stated in law, so any intelligent system will be able to suggest the maximal judgement in any case. Moreover. The above mentioned discussion referred to "Base penalty". However, the later paragraph (starting with "The convergence in the Figure 1") refers rather to "Final penalty" due to the invoked numbers. That really confuses the reader.

Ok, let us focus on the fact, that we have some data and we want to use them as the support for judges. We do not get anything practical from these tables, as they do not explain anything (I'll just remain the title of the paper: "Explaining the Judges' ") but only summarize previous sentences without any semantic context.

Honestly, I do not understand the idea of clustering judges due to the sentence time. First of all – why the number of clusters was set to 3? Was the set made to prove the already prepared thesis? That is not good to prove anything. But apart from that – only two clusters are described with the normal distribution.

But ok – let us follow that direction. The Authors provide no feature extraction from cluster related judgements to build any recommendation/explanation of already sentenced judgements. The only thing we get is a set of word clouds, that will be useless as a judge supporting tool.

What I was expecting – a text mining providing feature extraction and rules induction that would really help judges to set an appropriate and consistent decisions in a limited set of cases. What did we get – a chaos of possible ways of data processing without any consistent conclusions.

Of course, it is possible that Authors will state that I just did not understand the paper. Ok, I may agree, but that means, that the scientific core of the paper was not presented in an appropriate way. If it was not understandable for the reviewer, it may be not understandable for any other reader.

Apart from the above mentioned issues, there are also some minor (editorial and substantial) issues that should be taken into consideration when the improved version (or the new submission) would be prepared:

- I don't think that providing the formal notation in Introduction is a good practice. The introduction should announce the substantial content in the spoken language – not with the formalism. Especially, that here the notations were not clearly defined, as they should be in the scientific paper.
- Italic style: I can't guess what is the meaning of italic –shaped inclusions. A quote? We have the “” symbols for that (especially that they are used in a sentence starting from "As regards the legal nature "). An emphasis? I also do not buy it. A different language? Ok, that one application I would keep, while other should be presented in different but consistent way.
- Does the first name of “The Prompt Report: A Systematic Survey of Prompting Techniques.” really contain a ‘1’ in the end?
- A missing space after the end of the sentence starting "As regards the legal " and ending "just to cite some legal doctrines."
- Table 1 is not mentioned in the text.
- Table 2 location is too far from the first mention.
- Table 3. Why 0.33 (in years) is presented (in months) once as 4.00 (first line) and once as 3.99 (third line)? Just be consistent.
- Tables 4 to 7: shouldn't they have "samples" instead of "sample"? And they are not center as Table 3.
- I appreciate the Authors' will to present differences between US and Brazil law, however it could be just limited to one paragraph (and no table) as the paper does not focus on such differences. Moreover, it could be also moved into the Introduction as a justification to develop a Brazil law–dedicated system.

And for the end – once I've met the paper that was citing a song released on CD (I was the co–author of the paper) but it is a first time I met the Bible verse quote in the manuscript. With the decades of reviewing, it is still possible to surprise me ??

Concluding, that is not a bad paper, however it requires some efforts to make it a good paper. It is not so unusual that we start writing a manuscript with some idea and during the development and experiments the idea is evolving – that's the nature of science. However, it implies the necessity of verifying the original

idea (and the paper "working title") into the final results. I keep my fingers crossed that the paper will be accepted after modifications. In case of fail, I believe that You were given such an amount of constructive remarks, that next submission – wherever made – will end with success.

KDIR 2025 - Paper #76

Paper Title: *Explaining the Judges' Decisions Criteria*

Reviewer #2

General Assessment (Please assign scores using the following criteria (1=weakest; 6=strongest))

Relevance (Paper fits one or more of the topic areas?): 5
Originality (Newness of the ideas expressed): 4
Technical Quality (Theoretical soundness/methodology): 5
Significance (Is the problem worth the given attention?): 4
Presentation (Structure/Length/English): 5
Overall Rating (Weighted value of above items): 4

Improvement Suggestions (for authors to consider in the camera-ready version. Additional detail in "Observations")

Abstract and Introduction are adequate? Yes
Needs more experimental results? No
Needs comparative evaluation? Yes
Improve critical discussion ? (validation): No
Figures are adequate ? (in number and quality): No
Conclusions/Future Work are convincing? No
References are up-to-date and appropriate? Yes
Paper formatting needs adjustment? No
Improve English? No

Detailed comments to authors, including aspects that must be improved in the camera-ready version of the paper:

The paper focuses on the identification of named entities in judicial decisions to evaluate judges' reasoning and decision-making processes. By analyzing 769 judicial decisions from the S Paulo court, the research highlights differences in sentencing severity for crimes such as robbery, drug trafficking, and theft. It also examines how judges cluster based on their sentencing behavior, revealing anomalies linked to specific textual justifications. A key finding is the statistically significant difference in average sentences among the four crimes analyzed, with robbery and drug trafficking receiving higher sentences compared to theft. Although this finding is expected, its quantitative presentation is nonetheless interesting.

The methodology involves a standard application of a large language model, specifically the Qwen 2.5-32B-Instruct model. The study's contributions include making a large dataset of court decisions publicly available, organizing these decisions, and analyzing statistical similarities and discrepancies in judges' rulings.

While the overall approach is clear, it lacks direct quantitative benchmarking against prior methods, which is important for future studies. Additionally, I find Figure 5 unnecessary, as it resembles a cheap street advertisement and does not contribute meaningfully to the paper.

Overall, although the methodologies are standard and the results are expected, I still appreciated seeing how the authors carried out this work, which will be useful as a use case of LLMs for similar applications.

KDIR 2025 - Paper #76

Paper Title: *Explaining the Judges' Decisions Criteria*

Reviewer #3

General Assessment (Please assign scores using the following criteria (1=weakest; 6=strongest))

Relevance (Paper fits one or more of the topic areas?): 5
Originality (Newness of the ideas expressed): 5
Technical Quality (Theoretical soundness/methodology): 4
Significance (Is the problem worth the given attention?): 5
Presentation (Structure/Length/English): 4
Overall Rating (Weighted value of above items): 5

Improvement Suggestions (for authors to consider in the camera-ready version. Additional detail in "Observations")

Abstract and Introduction are adequate? Yes
Needs more experimental results? Yes
Needs comparative evaluation? Yes
Improve critical discussion ? (validation): Yes
Figures are adequate ? (in number and quality): Yes
Conclusions/Future Work are convincing? Yes
References are up-to-date and appropriate? Yes
Paper formatting needs adjustment? No
Improve English? No

Detailed comments to authors, including aspects that must be improved in the camera-ready version of the paper:

The paper tackles a socially and ethically relevant problem: how to interpret and explain judicial decision – making through data–driven methods. The proposed framework aims to uncover the latent criteria used by judges, by clustering decisions based on topic modeling and applying interpretable post–hoc techniques (like decision trees) to associate decisions with contextual features .

The originality of applying explainability techniques in the legal domain is notable. This aligns very well with current KDIR themes, particularly in Explainable AI and knowledge extraction from unstructured and sensitive textual data. The effort to ensure interpretability and transparency, rather than pure performance , is especially commendable in this context.

The approach is described clearly , and the structure of the pipeline from topic extraction , to decision classification , to tree–based explanation is coherent and replicable . However, certain components could be expanded or refined. For example, while LDA is a reasonable baseline for topic modeling, the increasing availability of transformer–based embeddings or BERTopic–like methods might provide richer semantic grouping. Similarly, the validation of the explanation quality (e.g., fidelity of the decision trees , alignment with expert opinion) could be elaborated.

From an experimental standpoint, the analysis on a real–world dataset is appreciated, but the study could

benefit from more comparative baselines and a deeper discussion of failure cases, bias, or generalizability .

Overall, this is a meaningful and promising paper. With a few refinements, it could become an important reference in the space of interpretable legal AI systems. The authors are encouraged to continue this line of work and to further investigate how such explanations might be received or validated by legal professionals .

KDIR 2025 - Paper #76

Paper Title: *Explaining the Judges' Decisions Criteria*

Reviewer #4

General Assessment (Please assign scores using the following criteria (1=weakest; 6=strongest))

Relevance (Paper fits one or more of the topic areas?): 4
Originality (Newness of the ideas expressed): 4
Technical Quality (Theoretical soundness/methodology): 3
Significance (Is the problem worth the given attention?): 4
Presentation (Structure/Length/English): 3
Overall Rating (Weighted value of above items): 4

Improvement Suggestions (for authors to consider in the camera-ready version. Additional detail in "Observations")

Abstract and Introduction are adequate? No
Needs more experimental results? No
Needs comparative evaluation? Yes
Improve critical discussion ? (validation): Yes
Figures are adequate ? (in number and quality): No
Conclusions/Future Work are convincing? No
References are up-to-date and appropriate? Yes
Paper formatting needs adjustment? Yes
Improve English? No

Detailed comments to authors, including aspects that must be improved in the camera-ready version of the paper:

I am border line on acceptance of this paper as even though the topic is interesting and there are some results that may be worthy of publication the way in which the paper is written and presented has a number of issues that make it less suitable for publication. Here are the issues that I find with it which the authors should really try to improve on. Problems are listed below (in the order of appearance, not of importance):

- 1) The abstract could be stronger in terms of presentation of results and should have the proverb removed.
- 2) Formatting is in places odd, e.g. the italics of specific sentences such as the one in page 1 column 2 and other places such as beginning of the lit . review.
- 3) Even though the introduction talks about areas of experimentation such a prompt engineering no results of this are formalised. This weakens the paper as there should be presentation of how something like prompt engineering, which is quite novel, applied in this case and some result presentation.
- 4) There is a limitation in the data collection which refers only to 2022. There could be evolution over time so that is rather restrictive and should be mentioned as a limitation of the dataset and justified .
- 5) The paragraph starting "Recent advances have pointed out on " is poorly written and needs revising.
- 6) I would question given the nature of the paper and the research undertaken whether there is a need to present things such as legal positivism theory and the whole of that paragraph given that it is not used in the paper.

- 7) Much of what is on the lit. survey should really be contained in a section called methods which is missing in the paper!
- 8) It is not clear at all why there is a need for such lengthy discussion on the differences between the Brazilian and American legal systems since that really is not the focus of the paper and there is no experimentation that follows that necessitates that.
- 9) There is a section called the name entities which does not present in itself very useful information as it does not really conclude for example what name entities were extracted and the results don't give any clues on this either.
- 10) The prompting engineering strategy section talks of various strategies used but then no results are presented on this. Again this is an omission as it could have been an interesting part of the paper if the presentation of results and discussion of those was included.
- 11) Not sure about the value of table 2 or where it is referred to in the text. That part of the paper does really seem superfluous and should have been used to better present results.
- 12) Results presentation should improve. For example why present first table 3 when the information is presented for that particular crime again later? Seems an intermediary step not really necessary. The explanation in detail of that table seems also superfluous as most people with a reasonable understanding of statistics will not need to be told the effect of particular values of mean, median etc.
- 13) Tables 4 to 7 should usefully be merged into one which would allow for better comparison and better presentation.
- 14) Much of what is discussed in the results (e.g. around ANOVA) should be in a methods section as not results.
- 15) Since the parameters of sentencing for each crime are different (e.g. min and max) one would clearly expect that there is going to be a difference in means. Not sure that is a very useful result.
- 16) Why use 3 clusters arbitrarily? There are methods to find what may be the right number of clusters and they should be used.
- 17) It is not quite clearly explained what variables were used in the clustering. This should be made clear. Is figure 3 relating to one crime type or all? Again not clearly presented.
- 18) Conclusions: not sure that the proverb there is useful or appropriate. Also the paragraph in italic with citation probably not appropriate. Conclusions are not discussion!
- 19) Conclusions should really be concrete and a summary of the strong findings. I think they need to be rewritten and then results be reflected in the abstract.
- 20) Referencing format is problematic. From references without enough detail (e.g. Austin with no details of publication) to some with so many names and no other details (e.g. Quen or Schulhoffl).

So overall it may be an interesting paper but it does need rewriting to be a good paper, taking out irrelevant detail, improving results, with a methods section and conclusions that are reflective of the findings.

KDIR 2025 - Paper #76

Paper Title: *Explaining the Judges' Decisions Criteria*

Reviewer #5

General Assessment (Please assign scores using the following criteria (1=weakest; 6=strongest))

Relevance (Paper fits one or more of the topic areas?): 4

Originality (Newness of the ideas expressed): 5

Technical Quality (Theoretical soundness/methodology): 5

Significance (Is the problem worth the given attention?): 5

Presentation (Structure/Length/English): 5

Overall Rating (Weighted value of above items): 5

Improvement Suggestions (for authors to consider in the camera-ready version. Additional detail in "Observations")

Abstract and Introduction are adequate? Yes

Needs more experimental results? No

Needs comparative evaluation? No

Improve critical discussion ? (validation): No

Figures are adequate ? (in number and quality): Yes

Conclusions/Future Work are convincing? Yes

References are up-to-date and appropriate? Yes

Paper formatting needs adjustment? No

Improve English? No

Detailed comments to authors, including aspects that must be improved in the camera-ready version of the paper:

The article is very well-written and addresses a highly relevant issue by attempting to objectify judicial decisions using robust clustering methods. The procedure used to analyze the corpus of court rulings is particularly solid, well-structured, and rigorously executed. This adds significant value to the study and provides meaningful insights into patterns of sentencing and judicial behavior.

However, I believe the paper should provide a more detailed explanation of how the privacy of all parties involved in the judicial process is protected. Court rulings often contain personal data such as the names of defendants, victims, and judges and it is not entirely clear how this information is anonymized or pseudonymized before applying clustering algorithms. This is especially important in the section where judges are clustered according to the sentences they issue for different types of crimes. As it currently stands, the methodology could allow for the identification of individual judges and potentially expose patterns that may lead to reputational, ethical, or safety concerns.

Explaining the Judges' Decisions Criteria

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Keywords: knowledge organization, clustering, large language models, information extraction, judicial decision

Abstract: The identification of named entities in free text is a foundational research area for building intelligent systems in text and document mining. These textual elements allow us to evaluate the reasoning expressed by document authors. In a judicial decision, for example, by identifying time-related entities, an intelligent system can assess and verify whether a sentence issued by a justice agent falls within socially agreed-upon statistical parameters. In this study, 769 judicial decisions from the São Paulo court were evaluated. Our experiments compared the extreme time-value sentences against those with the lowest sentence, for instance, to infer the expressions that justified and have explained their values. The results revealed differences in sentence severity among robbery, drug trafficking, and theft, as well as in how judges cluster based on their sentencing behavior. The study also highlights anomalies in sentencing and links them to specific textual justifications, demonstrating how judges' decisions can reflect both legal criteria and subjective biases. [...] *In a lawsuit the first to speak seems right, until someone comes forward and cross-examines.* (Proverbs 18:17)

1 INTRODUCTION

Information retrieval tools have significantly improved over the last decade. Companies like Google, Yahoo, and Bing provide powerful search engines that enable near-instantaneous access to vast amounts of documents worldwide. Nevertheless, we need to further improve these tools to provide the user with better search engine for more complex meta-indexing (Izo et al., 2021). Inasmuch as similar tools are still lacking in many information systems used by the Brazilian justice department. Integrating such tools into local institutional systems could both enhance efficiency by enabling precise and rapid retrieval of information from extensive document archives (Lima et al., 2024) and allow one to perform complex inferences on hitherto statistical parameters.

The judge's decision-making landscape must be minimally justified in their decision document. Therefore, to apply a sentence s_i , the judge must justify it by presenting facts $\mathcal{F}$, in other words, we must find these elements, $F_{acts} \rightarrow s_i$, explicitly registered in the decision. $\mathcal{F}$ can be either the defendant's past criminal history f_1 , or the defendant's ties to the community f_2 , or the defendant's employment history f_3 , or any other fact f_n , or all together, one may add to the decision's description. Therefore, $F_{acts} = \{f_1, f_2, \dots, f_n\}$, where $F_{acts} \subset \mathcal{F}$, are the explanation a judge gives to charge someone with a s_i penalty.

Judges can also adopt the strategy of comparing two, or more cases, to set up their decision. In this case, a threshold is used to say that F_i and F_j – where $F_i, F_j \subset \mathcal{F}$ – are somehow similar to induce the same sentence s_i . By doing so, judges' decisions become as rational and clear as mathematics, aligning with society's expectation that judges remain impartial and dispassionate (Maroney, 2011). Unfortunately, some judges do not decide entirely on the record. They add to the judgment decision their emotional feeling and

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sympathy instead of the logic and strict description of the law. Other studies, discussed by Rachlinski and Wistrich (2017), demonstrate that specific religious values lead judges to decide certain kinds of cases differently. Also, Rachlinski and Wistrich (2017) examined various cognitive biases that may influence judges when issuing their verdicts. Some are: (a) demographic characteristics of judges. (b) relying heavily on intuitive reasoning and not on facts, or even worse, relying on facts outside the judicial records. In Brazil, we may include the social class of the defendant as a bias factor on the judicial decisions. In theory, judges' beliefs should not override the original agreement expressed by the people in the ordinary laws established under the Brazilian Constitution.

The judge's decision is the final administrative act issued by the judicial system referring to facts disputed by parts. Having this in mind, these documents can be valuable instruments to assess the performance of the system. However, to achieve this goal, we need to build a better archive system to increase the accessibility of these documents by attaching them to new access points. Today's accesses are based mainly on the number of processes, the name of the parts, the subject type in dispute, and keywords.

The extraction of the aforementioned metadata may be straightforward using pattern recognition strategies. In this work, we go beyond that by extracting other information within these documents, such as the names of other people mentioned, the dates of the referenced events, and the time prescribed by the court. More complex pieces of information require Natural Language Processing (NLP) (Manning and Schuetze, 1999). To handle these more complex extractions, this work incorporates prompt engineering techniques, which have proven effective in guiding natural language models. This approach involves the careful formulation of instructions for Large Language Models (LLM)s, enabling a more efficient use of their capabilities. Understanding how to construct, refine, and strategically apply prompts is essential to maximizing these models' performance. Well-crafted prompts allow the model to recognize subtle patterns, such as temporal relationships and specific entities (Schulhoff1 et al., 2024). Thus, prompt engineering serves as a bridge between expert knowledge and artificial intelligence, expanding the practical application potential of these models.

The contributions of this work are threefold. First, we make publicly available a large dataset of court decisions regarding the crimes discussed here. Second, we organize these decisions into popular software frequently used by archivists, such as *AtoM*¹. Finally,

¹<https://www.accesstomemory.org/>

we analyze 769 decisions from 290 different equivalents to USA district judges. We attempted to explain their statistical similarities and discrepancies by extracting factual elements from their decision documents.

To present our proposal, we consider only four types of crime: 1) Simple Theft, 2) Qualified Theft, 3) Robbery and 4) Drug Trafficking. We formed an archive of 769 decisions from that issued by the *São Paulo Court of Appeal* – (TJSP) in 2022 only.

This article is structured as follows. In Section 2, we discuss some recent works that will serve as a basis for comparison with our proposal. In Section 3, we present our proposal with an emphasis on the efficient use of computational resources and artificial intelligence techniques aimed at the extraction and structuring of legal data. In Section 4, we discuss the experiments and results. Finally, we present our conclusions in Section 5.

2 LITERATURE REVIEW

Advances in artificial intelligence and data analytics have enabled computers to predict judicial decisions by analyzing patterns in case law, judges' rulings, and legal reasoning.

Regarding judicial decision-making and the application of technologies, the seminal Lawlor's approach, developed by legal scholar Reed C. Lawlor, arose in the 1960s. His work applied early computational logic to legal reasoning, using syllogistic structures to model how judges interpret facts and legal rules. Lawlor's method broke down judicial decisions into discrete components - facts, legal principles, and outcomes - to identify predictable decision pathways. From 70s to 90s, there were proposals to combine the automation of legal reasoning and expert systems, such as: TAXMAN (McCarthy, 1977) and FINDER (Tyree et al., 1987). Later, a computational breakthrough was introduced by Susskind (1998), and Ashley (1990) proposed a rule-based and case-based reasoning paradigm, respectively. *These theoretical debates set the stage for computational models aiming to replicate or predict judicial reasoning.*

A paradigm shift from symbolic AI to data-driven methods was proposed by Bench-Capon and Sartor (2003), emphasizing the value-driven reasoning in AI systems, and Katz et al. (2017) applying techniques such as machine learning, considering data-based decision-making but ignoring some *contextual factors*.

Recent advances have pointed out on interdisciplinary innovations, for instance, NLP and Bench-

marking Chalkidis et al. (2023), with *LexGLUE* dataset standardizing legal NLP tasks; and topics as ethical risks (Sourdin, 2021), such as the ethical concerns about replacing judges with algorithms; biases and hallucinations in automatic judicial decision making (Medvedeva et al. (2020): how evaluates biases in AI predictions of human rights cases, and explainability (how adapt AI for different legal systems).

In recent years, several projects have been suggested to study how technology can be used in courts. One of these projects is called ADELE. The ADELE project tried to create a plan for using technology to understand legal data. It focused on court decisions to help analyze and process legal data more effectively. The main goals were to enhance legal data analysis, improve accessibility to case law insights, and provide a structured approach to interpreting judicial decisions in these domains. The approaches applied were Natural Language Processing (NLP) and machine learning (ML) to analyze court decisions in Italian and Bulgarian case law, focusing on Intellectual Property Rights (IPR) and Value Added Tax (VAT). Key methods included text preprocessing, named entity recognition (NER), and topic modeling to extract legal concepts, classify cases, and identify patterns. The project also utilized predictive analytics and knowledge graphs to map legal relationships and precedents, enabling structured analysis of judicial decisions.

While these approaches offer valuable insights for legal practitioners and policymakers, challenges remain, such as ensuring algorithmic transparency and addressing biases in training data. Nevertheless, the integration of computational methods into judicial analysis represents a significant evolution in legal research, bridging traditional doctrinal analysis with data-driven forecasting.

As regards the legal nature of judicial decision, its definition varies across legal traditions, for instance, in Legal Positivism theory, a judicial decision is "the application of pre-existing rules to facts, deriving authority from the legal system's hierarchy of norms." (Hart, 1961); in Legal Realism theory, judicial decisions are influenced by contextual factors (e.g., judge's worldview, societal values) beyond formal rules (Fuller, 1958); in Legal Interpretivism, decisions encompass principles (e.g., justice, fairness) alongside rules, constructing a coherent moral narrative (Dworkin, 1977), just to cite some legal doctrines. *These theories reveal tensions between rule-bound formalism and discretionary judgment, shaping computational modeling attempts.*

The Brazilian penal legal system is grounded in a combination of legal theories (e.g., legal positivism,

social-critical critiques) that shape its substantive (*Direito Penal*) and procedural (*Direito Processual Penal*) dimensions. In this work, the scope of the Brazilian theory of judicial decisions was limited to sentences, which are a final court ruling that resolves the merits of a case. Interlocutory decisions or provisional orders were excluded from the scope.

A *Judicial Sentence* is issued by a competent judge, who articulates the factual and legal basis for the ruling. In terms of communicative acts, judicial sentences are a subtype of institutional acts (Searle, 1995; Austin, 1975), i.e., an act requiring objective and subjective validity conditions. Under the Brazilian Code of Criminal Procedure (*Código de Processo Penal – CPP, Art. 381 to 383*), to be considered as a valid judicial sentence (*sentença penal*) must be signed and dated, and necessarily containing 1) the identification of all parties involved in the case, including their full names or, when such information is unavailable, any other details sufficient to establish their identity; 2) must present a concise description (*aka relatório*) of both the prosecution's accusations and the defense's arguments, outlining the key claims and counterarguments that frame the legal dispute; 3) the legal reasoning (*aka fundamentação*), which is the factual analysis, the subsumption of the fact to the criminal type (analyzing materiality, culpability, legal grounds). For example, the defendant's nocturnal entry into the victim's property, captured on CCTV, satisfies the aggravator of 'night-time' under Art. 155, §4º; 4) the dispositive part (*aka dispositivo*), containing the final ruling, a clear decision on conviction or acquittal, the penalty in case of convicted, and, in some cases, civil liability. Also, there are some differences between criminal type definitions as shown in Table 1.

The legal definitions and treatment of theft, robbery, and drug trafficking reveal significant differences between Brazil's civil law system and the United States' common law tradition. These distinctions reflect broader variations in legal philosophy, legislative structure, and societal priorities.

In Brazil, simple theft is defined under Article 155 of the Penal Code as the unlawful taking of movable property without consent and without violence, carrying a penalty of one to four years imprisonment. The U.S. approach, guided by state statutes and the Model Penal Code, focuses instead on the value of stolen goods, distinguishing between petty theft (minor offenses) and grand theft (more serious crimes), with penalties varying accordingly. Brazilian law further recognizes qualified theft, where aggravating factors such as breach of trust or nighttime commission increase the sentence, while U.S. law lacks an exact

counterpart, instead enhancing penalties based on victim vulnerability or stolen property value.

Robbery in Brazil, as outlined in Article 157 of the Penal Code, requires theft accompanied by violence or grave threat, punishable by four to ten years in prison. The U.S. adopts a broader definition, where any use of force or intimidation during theft constitutes robbery, with penalties ranging from ten years to life imprisonment, particularly in cases involving firearms. This reflects a stricter stance on crimes involving personal violence in the U.S. compared to Brazil’s more narrowly defined thresholds for aggravated theft.

Drug trafficking presents another area of contrast. Brazilian law (Law 11.343/2006) criminalizes unauthorized production, transport, or sale of illicit drugs, with sentences of five to fifteen years, while distinguishing between personal use and trafficking based on quantity. The U.S., under federal law (21 U.S.C. § 841), emphasizes intent to distribute, imposing severe penalties—often five to forty years—depending on drug type and quantity, with mandatory minimum sentences playing a prominent role.

These differences highlight how Brazil’s codified legal system provides precise definitions and fixed sentencing ranges, whereas the U.S. system’s flexibility allows for contextual adjustments based on judicial interpretation and legislative enhancements. Understanding these distinctions is crucial for comparative legal studies and cross-jurisdictional policy analysis.

3 THE METHODOLOGY

A judge’s decision statement contains numerous facts, F_{acts} , that may hold valuable information for someone in the future. However, it is impossible to predict *a priori* which words, complex textual structures, or other linguistic phenomena will be most useful to a researcher, citizen, or archivist as their access points. Therefore, we need enhanced tools and methodologies to efficiently adapt archival description structures to meet users’ specific needs. The case study proposed in this work is the evaluation of the behavior of judges on sentencing. This information was not previously available to the information system for us to process and perform various statistical analyses. For that reason, we needed to extract them from the documents using modern artificial intelligence tools.

To protect the privacy of all parties involved in the analyzed judicial documents, we adopted a rigorous anonymization process. All personal information – including names of defendants, victims, wit-

nesses, and judges – was removed or replaced with coded identifiers before applying the clustering algorithms. Additionally, we ensured that the identifiers used do not allow reverse re-identification, even through cross-referencing analysis. This careful approach is especially important in grouping judges by sentencing patterns, where we take additional measures to prevent any risk of individual exposure.

3.1 The prompting engineering strategy

To construct the database, structured information was automatically extracted from judicial rulings, including case number, category, district, court division, presiding judge, imposed sentences (base, provisional, and final), as well as judicial circumstances and aggravating or mitigating factors such as recidivism, use of weapons, attempt, voluntary confession, and characteristics of both the defendant and the victim. The extraction process was carried out with the assistance of the Qwen 2.5-32B-Instruct language model (Qwen et al., 2025), an open-source model with 32 billion parameters, run locally. The choice of this dense version of the model ensured greater control over processing, preservation of the confidentiality of judicial decisions, and advanced performance in tasks involving legal language understanding and structured data handling. A prompt was developed to traverse all decisions, identifying sentences and correctly linking them to the respective individuals, as some rulings involve multiple defendants. This step was essential to properly structure the data, enabling subsequent analyses of judicial behavior in sentencing decisions.

The prompt followed a zero-shot approach, in which the model is guided solely by natural language instructions without being provided with prior input-output examples (Boyina et al., 2024). This technique allows for quick adaptation to new texts and tasks, although it may face limitations in more complex or ambiguous contexts, as is often the case with judicial rulings. For this reason, few-shot variations were also tested, incorporating concrete examples to guide the model regarding the expected format and the type of information to be extracted (Huang and Wang, 2025). The combination of these approaches proved effective in increasing the accuracy and consistency of the extraction process, contributing to the creation of a reliable and reusable dataset for empirical analyses of the criminal justice system.

Table 1: Comparative analysis of theft, robbery, and drug trafficking: Brazil vs. U.S.

Crime Type	Brazilian Law	U.S. Law	Key Distinctions
Simple Theft	Definition: Unlawful taking of movable property without violence or consent (Art. 155, CP). Penalty: 1–4 years + fine.	Definition: Unlawful taking with intent to deprive the owner permanently (no force). Penalty: Petty theft (≤ 6 months) to grand theft (1–3 years, state-dependent).	Brazil focuses on property type (movable), U.S. prioritizes property value thresholds.
Qualified Theft	Definition: Theft + codified aggravators (e.g., breach of trust, nighttime; Art. 155, §4°). Penalty: 2–8 years.	No standalone category; aggravators (e.g., victim age, high value) elevate to felony theft.	Brazil’s aggravators are explicit in law, U.S. uses contextual enhancements.
Robbery	Definition: Theft with violence/grave threat (Art. 157, CP). Penalty: 4–10 years (increased if armed).	Definition: Theft by force/intimidation (e.g., armed robbery; 18 U.S.C. § 1951). Penalty: 10 years–life (varies by state).	Brazil requires grave threat; U.S. accepts any force/intimidation.
Drug Trafficking	Definition: Unauthorized production/sale of drugs (Law 11.343/2006, Art. 33). Penalty: 5–15 years.	Definition: Manufacture/distribution with intent to distribute (21 U.S.C. § 841). Penalty: 5–40 years (mandatory minimums apply).	Brazil distinguishes trafficking from personal use by quantity. U.S. focuses on proven intent.

4 EXPERIMENTS AND RESULTS

The experiments carried out in this work show the behavior of a set of district judges from São Paulo, the biggest city in Brazil, with regard of four different type of crimes (see Section 2). Initially, we evaluated the best statistical distribution to characterize the Simple Theft sample of 99 sentences. In Table 2, we show the statistics for these sample sentences issued in our dataset for that type of crime.

Table 2: The simple theft statistics

	Simple Theft			
	Base Penalty		Final Penalty	
	year	month	year	month
minimum	1.00	12.00	0.33	4.00
average	1.20	14.37	1.16	13.96
σ	0.33	3.99	0.58	6.95
median	1.17	14.00	1.17	14.00
maximum	3.00	36.00	5.00	60.00

Based on the facts presented by the state’s attorney and the defendant, the judge may determine a *base penalty* in their decision (refer to the first and second columns in Table 2). This base penalty serves as the starting point, while the *final penalty* (see the third and fourth columns) results from adjustments – either increases or reductions – based on the evaluation of previously described factors. The values in the second column indicate that the minimum sentence is one year (365 days), while the maximum is three years. The average lies close to the center of this in-

terval. Adding further statistical insight, the standard deviation ($\sigma = 0.33$) is less than one year, indicating that for this type of crime, the variation around the average sentencing is relatively small. The third column shows similar figures but in months, considering all months with 30 days. These are two usual ways for professionals in the justice domain to deal with sentences. Above all, one can notice that all the median figures among year and month columns are very similar to the average values, suggesting a symmetric normal-curve.

Besides the previous deterministic experiments, we also carried out a different approach to estimate the same value discussed in Table 2 by applying the Probabilistic Programming (PP) technique with Stan language (Carpenter et al., 2017). PP is an approximation strategy, therefore the values of average and σ are not exactly the same of that calculated by the deterministic strategies in Table 2. Currently built within many modern languages, such as Python and R, PP uses an efficient engine to sample and re-sample to find the converging parameters to find a statistical distribution model for a given problem (Gelman et al., 2013). Nevertheless, we show the convergence of the approximated averages, for years, using a set of different samples in Figure 1.

The convergence in Figure 1 pointed out the average estimated by our PP approach. The approximated average given by the PP tool is 1.16, and the standard deviation is 0.59 at the level of the 65 sample items, much similar to that provided by the deterministic procedure in Table 2. Besides, note that the cred-

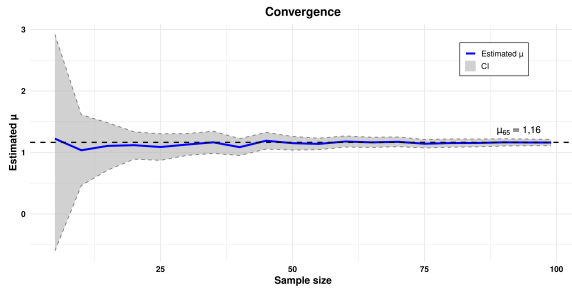


Figure 1: Convergence of years average for Simple Theft

ible interval shown in Figure 1 by the light gray area around the convergence line is narrowing as we increase the sample. This straight light gray area tells us we do not need a larger sample than 99 random sentences to conclude the average is 1.16 with 95% certainty.

The following Table 3 up to Table 6 present statistics for all four crimes considered in this study, each with a sufficient sample size to ensure that the average falls within a 95% credible interval.

Table 3: The simple theft statistics

Simple Theft – 276 sample					
	Base Penalty		Final Penalty		
	year	month	year	month	
minimum	0.29	3.50	0.22	2.67	
average	1.62	19.48	1.68	20.16	
σ	0.69	8.31	1.00	12.03	
median	1.25	15.00	1.33	16.00	
maximum	4.67	56.00	5.83	70.00	

Table 4: The qualified theft statistics

Qualified Theft – 174 sample					
	Base Penalty		Final Penalty		
	year	month	year	month	
minimum	1.00	12.00	0.67	8.00	
average	2.18	26.15	2.31	27.68	
σ	0.65	7.78	1.13	13.59	
median	2.33	28.00	2.04	24.43	
maximum	4.75	57.00	7.58	91.00	

Each table describes a model to explain mathematically its respective crime sentencing behavior by the population of judges of that district. With these *prior models*, we can compare and contrast judicial patterns across different districts and states in Brazil. In addition, this framework allows for discussions on whether these models reflect local culture and social norms or whether interventions might be necessary to influence or redirect judicial behavior.

A particularly interesting statistical analysis in this context is the analysis of variance (ANOVA),

Table 5: The robbery statistics

Robbery – 149 sample					
	Base Penalty		Final Penalty		
	year	month	year	month	
minimum	1.00	12.00	0.58	7.00	
average	4.57	54.82	7.52	90.25	
σ	0.78	9.35	4.24	50.90	
median	4.67	56.00	6.67	80.00	
maximum	6.67	80.00	36.00	432.00	

Table 6: The drug trafficking statistics

Drug Trafficking – 170 sample					
	Base Penalty		Final Penalty		
	year	month	year	month	
minimum	2.00	24.00	1.62	19.43	
average	5.48	65.77	4.63	55.55	
σ	0.87	10.41	2.47	29.69	
median	5.00	60.00	1.33	16.00	
maximum	8.00	96.00	9.72	116.67	

which revealed significant differences in the average sentences among the crimes evaluated (simple theft, qualified theft, robbery, and drug trafficking), with an F distribution value of 144.14 and $p < 0.0001$. This result shows that at least one group differs from the others. Tukey's *post hoc* test revealed that sentences for robbery are significantly higher than for the other crimes, and that drug trafficking also has higher average sentences compared to both simple and qualified theft. The only comparison without a significant difference was between simple theft and qualified theft, indicating that, in terms of average sentencing, these two types of theft are treated similarly in the dataset analyzed.

These results indicate a clear gradation in the severity of sentences, with robbery receiving the highest penalties, followed by drug trafficking, and finally theft-related offenses. Robbery stands out not only for having higher average sentences but also for greater variability, including extreme cases with very long sentences. Drug trafficking also shows a high median and greater dispersion compared to the theft categories. In contrast, simple theft and qualified theft exhibit similar patterns, with close medians and lower variability, reinforcing the lack of significant difference between them identified in the statistical analysis.

Note that within this dataset, there are 290 judges who, through their sentencing behavior, contributed to the models described by the parameters in the previous tables. In Figure 2, three clusters were defined with the aim of identifying patterns of similarity in sentencing, considering only those judges who sen-

tenced this type of crime. The choice of the number of clusters was guided by quantitative criteria, extracted during the automated clustering process. The K-Means algorithm was used, with its configuration driven by multiple internal validation indices, such as the Silhouette Score (SS), the Calinski-Harabasz index (CHS), and the Sum of Squared Errors (SSE). The combination of these indicators allowed identifying the three-cluster configuration as the most consistent in terms of group separation and internal cohesion of the data. Thus, the resulting structure reflects clusters based on objective quality metrics.

In Figure 2, the $N(\mu_{ST_l}, \sigma_{ST_l}) = N_{ST_l}(1.08, 0.37)$ normal distribution is a model to describe the first group of judges on the left. On the right, $N(\mu_{ST_r}, \sigma_{ST_r}) = N_{ST_r}(4.72, 0.58)$ explains the second group. Each model may account for the sentencing patterns observed among a particular group of judges in relation to this type of crime. The range of values of sentencing by these judges shows, hopefully, how different the circumstances faced by these judges for sentencing they did. Nonetheless, we are not yet looking into their sentencing text to combine the implication of their decision text and the time assigned to the sentence. We expect that the similar decision text should lead to the same time assigned to the sentence. These models also allow us to identify an unknown judge by one of these patterns by testing the judge characteristics under the null hypothesis, to a previous known $N_{ST}(\mu, \sigma)$. To build the models in Figure 2, we clustered all sentences' values in years and fractions because of months and days. That resulted in 51 decisions in the left cluster and 7 in the right cluster out of 276 decisions in total of this type of crime. The other sentences are in the clusters between these extreme ones.

So far, we illustrated our constructed mathematical structure only with $N_{ST}(\mu, \sigma)$ models, for simple theft. However, we have also built similar structures for: (a) $N_{QT}(\mu, \sigma)$, qualified theft; (b) $N_R(\mu, \sigma)$, robbery; and (c) $N_T(\mu, \sigma)$, drug trafficking.

The devised structures allow us to quickly search for the seven sentences, but one is overlapping another, placed on the right in Figure 2 as potential outliers sentences of this type of crime. We can also identify the judge, or judges, who have issued the court sentence in the $N_{ST_r}(\mu, \sigma)$ cases. Only six different judges issued those sentences, as shown in Figure 3. Among them, other twenty judges also issued sentences in different clusters. This unveils a very particular pattern of sentencing of these judges, which we can now look into the reasons why that occurred.

Similarly, we identified some outlier sentences for qualified theft in Figure 3. Using the same quantity of

four clusters previously applied to simple theft, four judges handed down sentences in the highest time-value group, while the remaining twenty were distributed among the other clusters.

To sum up, although the judges' decision texts differed from one another – explaining the variation in sentencing durations – some decisions appeared out of place or inconsistent with their respective clusters, as depicted in Figure 4.

We note that sentence 0, currently in Cluster 2, shares 95% similarity with Cluster 2. Similarly, sentence 1 shows 99% similarity with the same cluster. This suggests that both sentences could reasonably belong to Cluster 1, but are assigned to Cluster 2 due to their penalty values. Likewise, sentences 4 and 5 show strong similarity to those sentences in Cluster 1 – 100% and 97%, respectively. These sentences are considered outliers for this type of crime. Similar cases are also found on another type of crimes.

5 CONCLUSION

This work proposes the use of LLMs to extract complex information from legal texts, optimizing both archival description and the progress of legal proceedings through automation and semantic analysis. Unlike traditional approaches that focus only on elements such as case numbers or involved parties, our method analyzes the full text to identify named entities, allowing inferences about judges' behavior in their decisions.

The analysis focused on the duration of sentences to identify behavioral patterns, particularly in extreme cases. Empirical studies (Rachlinski and Wistrich, 2017) show that judicial decisions may be influenced by subjective factors—such as stereotypes or emotional states—rather than strictly legal reasoning.

One of the key findings is the connection between Figures 2, 3, and 4, all related to the crime of simple theft. Figure 2 presents the distribution of sentences across three distinct clusters, based on duration. Figure 3 retains the same sentences, now grouped by the judges who issued them, with colors preserved to represent the same clusters as in Figure 2. Figure 4, in turn, focuses on the sentences originally classified under cluster 2, redistributing them among the same three clusters—based on the textual similarity of each sentence to the centroids of these groups—while maintaining the same colors for comparison. This semantic comparison allows the user not only to verify thematic compatibility between the sentence and the reference group, but also to justify the inferred sentence based on observed linguistic patterns. The anal-

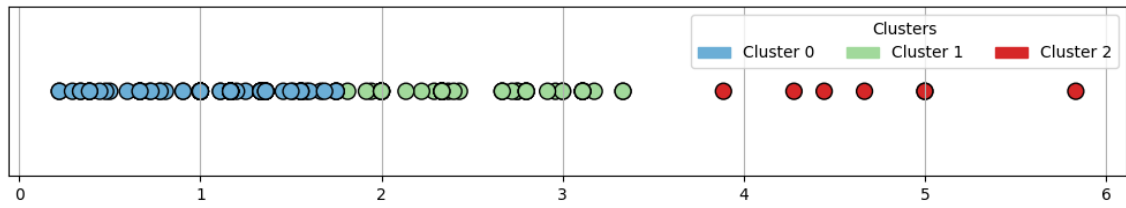


Figure 2: Clustering judges based on their sentencing to time – Simple Theft

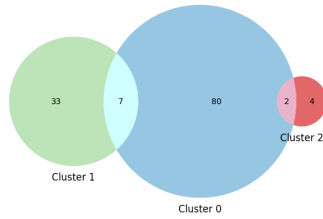


Figure 3: Simple Theft

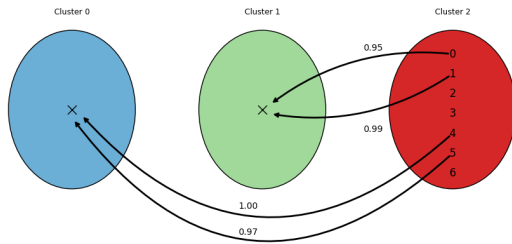


Figure 4: Clusters according to their texts – Simple Theft

ysis reveals that some decisions, although grouped by sentence length, are linguistically closer to the discourses of other clusters. This suggests inconsistencies in reasoning and possible influence of extralegal criteria in sentencing, even in legally homogeneous cases.

The proposed tool proved effective in detecting linguistic patterns and behavioral anomalies. However, discrepancies were observed between the terms identified automatically and those recognized by legal experts. As future work, we propose the development of ontologies to define metadata types and synonyms, the segmentation of clusters into more homogeneous subgroups, and the filtering of sentences with low semantic cohesion. We also suggest using embeddings or legal dictionaries to normalize linguistic variations and reduce noise in the analyses.

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